

Draft - expert review pending

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Topic 1.1 - What environmental science studies, and the planet it studies

A real classroom situation

Miss Farzana teaches Class 8 in a government girls' school in Karachi. It is November, and the city's air has turned grey. Before the annual sports day, she checks the air quality index, or AQI, on her phone the way other teachers check the weather. The reading is "unhealthy". She moves the races to the morning, shortens them, and tells the girls with asthma to sit out the long-distance events. A colleague asks her, half joking, "Since when are you a scientist?" Miss Farzana answers, "Since the air started deciding whether my pupils can run."

Her decision used a measurement, a standard, a health effect, and a judgement. That is environmental science in one classroom moment. This topic defines the field, maps its scope, and describes the planet it studies.

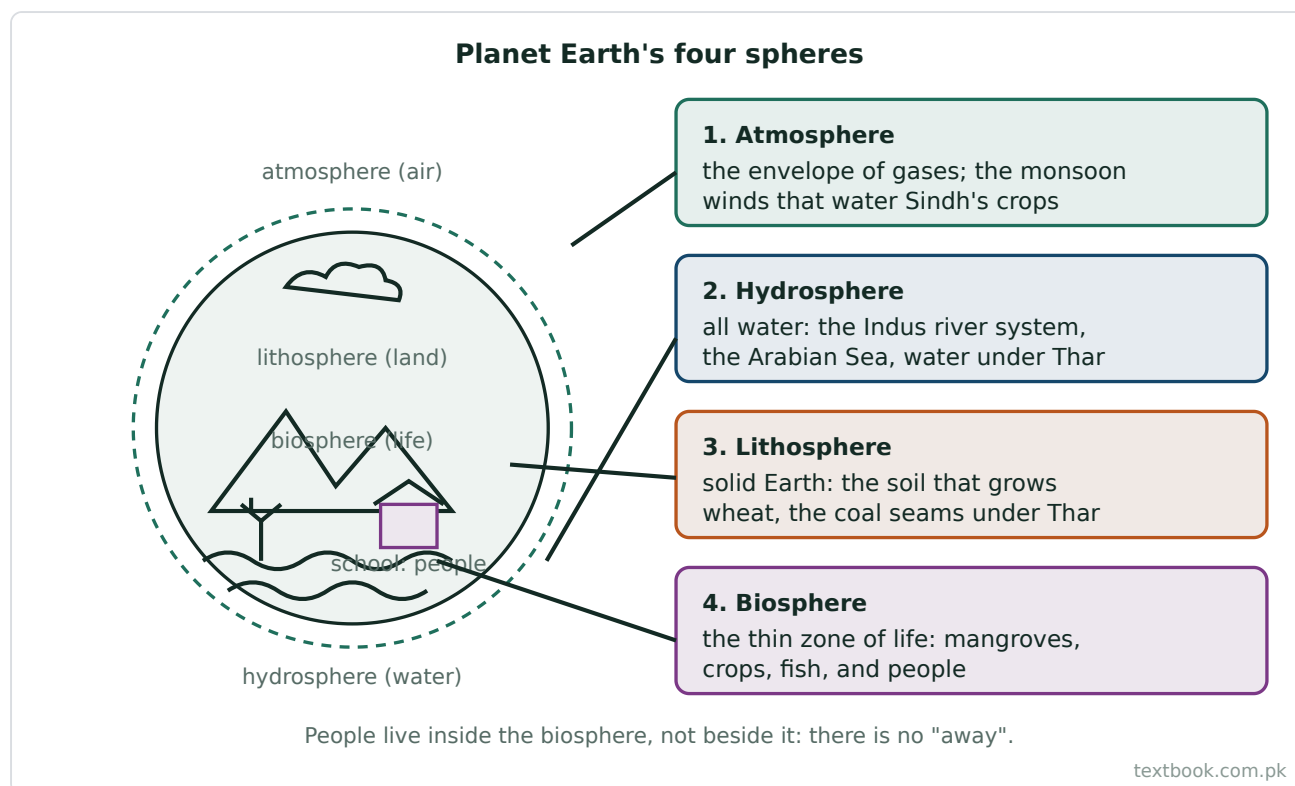
Explanation

What environmental science is, and why it matters

Environmental science is the study of the natural environment and the way people interact with it. It asks three linked questions: what is in our surroundings, how do human activities change those surroundings, and how do

the changes come back to affect people? It is not one discipline but a meeting point of several: chemistry for pollutants, biology for living things, physics for energy and climate, geography for where things happen, and the social sciences for why people act as they do.

The scope of the subject is easiest to see as five areas of questioning. The table below shows each area with one question a teacher in Sindh might actually ask.



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The importance of the subject is practical. Pakistan faces real environmental pressures: city air that repeatedly crosses unhealthy levels, a river system under strain, and a climate that is warming. A teacher who understands the subject can read an AQI number, explain why a canal smells, and answer pupils' questions with evidence instead of guesses. The World Health Organization reports that almost the whole world's population, 99 per cent, breathes air that exceeds its guideline levels (WHO, 2024). That single fact explains why this course exists.

A note on the word itself: the natural environment means everything around us that is not made by people: air, water, soil, rocks, plants, animals and the climate, together with the processes that link them. A classroom is not part of the natural environment, but the air its windows open onto is.

The natural environment and the structure of planet Earth

Scientists describe the natural environment as four great spheres that meet and overlap.

- The **atmosphere** is the envelope of gases around the Earth. Its lowest layer, where weather happens, holds the air we breathe and the pollutants we add to it. The monsoon winds that water Sindh's crops are atmosphere at work.
- The **hydrosphere** is all the water: oceans, rivers, lakes, groundwater, glaciers and ice. The Indus river system, the Arabian Sea and the water under the Thar desert are all hydrosphere.
- The **lithosphere** is the solid outer part of the Earth: rocks, soil and the landforms they make. The soil that grows Sindh's wheat and the coal seams beneath Thar sit in the lithosphere.
- The biosphere is the thin zone where life exists, where the other three spheres meet: the mangrove forests of the Indus Delta, the fish in Keenjhar Lake, the crops in the fields, and the people in the cities.

Beneath the biosphere, the Earth itself is layered: a thin crust, a thick mantle, and a core of iron. The crust matters most for this course, because that is where soil forms, water is stored, minerals sit and everything we build stands. When we speak of "using natural resources", we almost always mean taking something from the crust, the hydrosphere or the biosphere faster than it can be replaced.

What environmental science studies: five scope areas

Scope area	The question it asks	One Sindh classroom example
Air	what is in the air we breathe, and what did we add to it?	reading the AQI before a Karachi school sports day
Water	where does the water come from, and what is in it when it returns?	tracing the school tap back to the Indus or the ground
Land	how is the soil and rock used, and what covers them now?	the white salt crust on a waterlogged Sindh field
Living things	which species live here, and what happens when their habitat changes?	fish and birds of Keenjhar Lake
Human systems	how do laws, prices, cities and habits drive all of the above?	the price of a water bottle versus a filled bottle

One object, four spheres, one human system: the analytical habit of Topic 1.1.

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A worked example. Place a cup of tea on the desk. The water came from the hydrosphere, boiled by gas from the lithosphere. The tea leaf grew in the biosphere, in soil weathered from the lithosphere, watered by the hydrosphere, under the atmosphere's rain. The cup, the gas stove and the price of the tea belong to human systems, which environmental science also studies, because people are part of the environment, not visitors to it. One object, four spheres, one human system. That is the analytical habit this topic wants you to build.

Common misconception. Many people think "the environment" means nature far away from people: mountains, forests, wild animals. The four-sphere picture corrects this. People live inside the biosphere, drink the hydrosphere, farm the lithosphere and breathe the atmosphere. There is no "away". A city street is as much an environment as a mangrove creek, and the subjects of this course, from air quality to waste, are mostly about the environments people live in every day.

Activity: Sort the questions

Work in pairs. Your teacher will give you eight short newspaper headlines from a Pakistani paper, for example "Smog closes Lahore schools", "Indus flows low at

Kotri", "New coal plant approved", "Fish deaths reported at Keenjhar". For each headline, decide two things: which scope area it belongs to (air, water, land, living things, human systems), and which spheres are involved. Be ready to defend one answer that surprised you. Allow 20 minutes.

Check your understanding

1. Define environmental science in one sentence, in your own words.
(Remember)
2. Name the four spheres and give one example of each from your own district. (Understand)
3. A pupil says, "Environmental science is just about pollution." Give two reasons why this is too narrow. (Understand)
4. Place these three things in their sphere: the Indus river, Thar coal, the Indus Delta mangroves. Then explain why the mangroves belong to more than one sphere. (Apply)

Summary

Environmental science studies the natural environment and the human-environment interaction across five scope areas: air, water, land, living things and human systems. The natural environment is organised into four overlapping spheres: atmosphere, hydrosphere, lithosphere and biosphere, with people living inside the biosphere. The Earth beneath is layered into crust, mantle and core, and the crust is where most resources sit. The subject matters in Pakistan because its pressures, unhealthy city air, a strained river system and a warming climate, are already inside every classroom.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define environmental science and name its five scope areas.
- ☐ I can name the four spheres and give a Pakistani example of each.
- ☐ I can explain why people are part of the biosphere, not separate from it.

☐ I can take one everyday object and trace it through the four spheres.

Try this at your practicum school

Walk once around the school building with a notebook. Find one example of each sphere inside or beside the school grounds: the air over the courtyard, the water source or water tank, the soil in the garden or the dust on the windowsill, and any living thing from a tree to a sparrow. Write four sentences, one per sphere, naming what you found and one human activity that affects it. Do not name the school or any pupil.

Summative task

Write 150 to 200 words on the following.

A colleague tells you, "The environment is not our department. We teach, nature looks after itself." Using this topic's account of environmental science, the four spheres and the scope table, explain what the colleague is missing. Support your explanation with at least two examples from Pakistan.

Mini-rubric

Criterion	What a strong response does
Definition used (Understand)	States what environmental science studies rather than only defending it
Sphere reasoning (Analyze)	Traces at least two examples through the spheres to show people are inside the environment, not beside it
Pakistani evidence (Apply)	Uses specific, real examples (city air, the Indus, Thar) rather than generic "pollution"
Clarity and register	Plain sentences for a colleague; within the word count

Further reading

- World Health Organization. (2025). *Biodiversity* (fact sheet). The biosphere as the foundation of human health: <https://www.who.int/news-room/fact-sheets/detail/biodiversity-and-health>
- Haines, A., & Frumkin, H. (Eds.). *Planetary Health*. Island Press. (guide-recommended book, cited by reference only)

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